

This is sample data provided for CALIBER purposes only		INTERLOCK LOGIC DIAGRAM & CAUSE / EFFECT MATRIX		DOC NO: TJC-LLD-IL-FA-8901 WORK NO: BA-0256 REV: 3	
LOGIC No: SEQ-8901		DESCRIPTION: REFLUX ACCUMULATOR (FA-8901) OVERHEAD PROTECTION LOGIC			SIL: SIL 1 TAG: FA-8901 FLOC: TJC-LLD-8900-01

CAUSE & EFFECT MATRIX

ID	INITIATOR / CAUSE (INPUT)	TAG	SET POINT	VOTE	EFF-1 TRIP COLUMN FEED (prevent carryover)	EFF-2 TRIP REFLUX PUMP GA-8920 (protect)	EFF-3 OPEN VENT PCV-8905 (relieve pressure)	EFF-4 ANNUNCIATE ALARM ON DCS
T1	Drum level HIGH-HIGH	LSHH-8901	> 85 %	1oo2	X			X
T2	Drum level LOW-LOW	LSLL-8901	< 15 %	1oo2		X		X
T3	Drum pressure HIGH-HIGH	PSHH-8902	> 8 barg	1oo2			X	X
T4	Emergency stop (manual)	HS-8901	Manual	1oo1	X	X	X	X

Legend: X = initiator (row) actuates that effect (column). Voting: 1oo1/1oo2/2oo3 = number of transmitters that must agree to trip.

EFFECT ID	FINAL ELEMENT / ACTION
EFF-1	TRIP COLUMN FEED (prevent carryover)
EFF-2	TRIP REFLUX PUMP GA-8920 (protect)
EFF-3	OPEN VENT PCV-8905 (relieve pressure)
EFF-4	ANNUNCIATE ALARM ON DCS

#	START PERMISSIVE (AND-gate)	SIGNAL
1	Level in normal band (20-80%)	LT-8901
2	N2 blanket healthy	PT-8902
3	Boot not flooded	LT-8903
4	No active trip / reset done	DCS reset
=>	ALL permissives TRUE (AND) => START FA-8901 ENABLED	RUN

NOTES: 1. Trip set points are DUMMY training values. 2. On any trip the effects marked X are actuated by the Safety PLC and annunciated on the DCS. 3. A trip is latched and requires a manual reset once the cause has cleared and permissives are healthy. 4. Refer to P&ID; TJC-LLD-PID-8901 for instrument loops and the Datasheet for design limits of FA-8901.